GURURAGHURAMAN SETHURAMAN

Plot No 2C, Gurushika, Lakshitha Blooms, Bala garden extn., Noothencheri, Madambakkam, Chennai - 600126
Ph: +91 7395971667 ; E-mail: s.gururaghuraman@gmail.com

ACADEMIC QUALIFICATION

- **Bachelor of Technology in Electrical and Electronics Engineering**, Amrita Vishwa Vidyapeetham, Coimbatore, India; CGPA: 7.87/10; May’25
- **Erasmus Exchange Program in Nanotechnology**, Grenoble-INP PHELMA, Grenoble, France

WORK EXPERIENCE

Database Architect & Python Developer, **AInsurCo, London, The United Kingdom (Remote)** ***Aug’25–Present***

An insurance technology company developing software solutions for insurance operations, including data management, process automation, and digital insurance workflows.

- Designed and maintained 10+ relational database schemas/tables supporting core insurance data operations, applying normalization and indexing on frequently queried and joined columns to improve data retrieval performance
- Wrote and optimized 30+ SQL queries and stored procedures for reporting, validation, and data-retrieval workflows across the platform
- Resolved data-integrity and schema-consistency issues through constraint enforcement and automated validation checks
- Built 5+ Python automation scripts supporting scheduled batch processing and database workflows

Embedded Hardware Intern, **Robert BOSCH Limited, Bangalore, India** ***Jun’24 - Jul’24***

Global engineering and technology company operating across mobility, industrial technology, consumer goods, and energy and building technology, with significant expertise in automotive systems, electronics, and vehicle electrification.

- Developed and validated 5+ Embedded C firmware modules implementing CAN communication layers for EV benchmarking infrastructure and VCU interfaces
- Analyzed 100+ CAN frames using CANalyzer, performing frame-level logging, DLC verification, error-state capture, and fault-injection analysis across 3+ live HV testbeds
- Contributed to technical evaluation of E-axle architecture across 2 HV subsystem configurations, documenting integration behavior per internal test protocols

Supply Chain Intern (Electronics Division), **Schneider Electric, Chennai, India** ***Jul’23***

An energy management and industrial automation company providing technologies and solutions for electrification, automation, digitalization, and energy efficiency across industrial and commercial applications.

- Validated 150+ PCB components across power-electronics manufacturing lines, identifying non-conformances against design specifications
- Analyzed component lifecycle and supply dependencies for 8+ critical converter subsystems, supporting procurement risk assessment

Manufacturing and Quality Intern, **Igarashi Motor Pvt Ltd, Chennai, India** ***Feb’23***

Automotive component manufacturer specializing in precision electric motors and electromechanical systems for automotive and industrial applications, with a strong focus on manufacturing and quality engineering.

- Audited motor assembly workflows across 3+ production stations, identifying redundant checkpoints and opportunities to improve inspection efficiency
- Reviewed and improved BOM accuracy for 20+ line items, strengthening defect-traceability documentation for small-motor production

Quality Intern, **Sri Balaji Assembles and Plastics India Pvt Ltd, Chennai, India** ***Jul’22 - Aug’22***

Manufacturing company involved in plastic component production and assembly, supporting automotive and industrial applications through manufacturing, assembly, and quality processes

- Supported IPC/ISO-compliant quality control checks across 50+ electronic sub-assembly units
- Collaborated with production teams to isolate component-level defects across 10+ assembly batches, reducing rework cycles

ACADEMIC PROJECTS

Title: Li-Ion Battery Protection PCB
Duration: May’26 - Jul’26

Team size: 1
Summary: Designed and simulated a 2-layer Li-Ion battery protection PCB using KiCad 9 and LTspice, integrating reverse-current protection, a 500 mA resettable PTC, and ADC-compatible battery-voltage monitoring. Completed circuit simulation and manual PCB layout for an 11-component, 90 × 40 mm board with ground pours and design-for-manufacturing considerations. Passed DRC with zero violations and generated fabrication-ready Gerber and Excellon files.

Title: MTJ-based Random Number Generator
Duration: Feb'25 – May'25
Team size: 6
Summary: Developed a Python-based stochastic Random Number Generator using Magnetic Tunnel Junctions (MTJs), modeling probabilistic switching behavior as a physical entropy source. Implemented simulation workflows and conducted 25+ trials to evaluate statistical randomness and RNG performance. Investigated MTJ-based random generation for low-power computing and spintronic devices.
Individual Contribution: Curated the numerical simulation workflow and implemented the MTJ stochastic switching model; investigated and resolved implementation issues through independent study of the underlying physics and numerical methods.

Title: Demagnetization Fault Prediction in IPMSM using ML
Duration: Jun'24 – Apr'25
Team size: 4
Summary: Developed a machine-learning-based fault-diagnosis framework using FEM-simulated IPMSM datasets generated in ANSYS Motor-CAD across 10+ demagnetization severity levels. Extracted statistical features from phase-current signals and evaluated Gradient Boosting, Random Forest, SVR, KNN, and Linear Regression models using MAE, MSE, and R^2. Built a reproducible Python/Scikit-learn pipeline for demagnetization-severity prediction and healthy/faulty classification.
Individual Contribution: Spearheaded FEM simulation and dataset generation, feature extraction, and model development; coordinated a four-member team and contributed to research papers presented at IEEE conferences.

PUBLICATIONS
S. Guru Raghu Raman, S. Hari Venkatesh, R. Manuprabha, H. Renill and N. Praveen Kumar, "Demagnetization in V-shaped IPMSM: A Fusion of Finite Element Analysis and Machine Learning," published in 2025 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT), Bengaluru, India, Jul'25, pp. 1-6, doi: 10.1109/CONECCT65861.2025.11306357.

AWARDS & ACHIEVEMENTS
Best Paper Award for the paper titled 'Demagnetization in V-shaped IPMSM: A Fusion of Finite Element Analysis and Machine Learning', selected among 12 papers in the session, by IEEE CONECCT, Bengaluru, Jul'25

WORKSHOPS
Participated in a 1-day workshop on 'Xilinx Vivado – FPGA Board', conducted by the Department of Electrical and Electronics Engineering, held at Amrita Vishwa Vidyapeetham, Coimbatore, Sep'24

TECHNICAL SKILLS
- Languages: C, Embedded C, Verilog, Python, SQL
- Microcontrollers & Platforms: STM32, dsPIC, ESP32, Arduino, Raspberry Pi
- Protocols: CAN, UART, SPI, I2C
- EDA & Simulation: MATLAB/Simulink, ANSYS Motor-CAD, Vivado, KiCad 9, LTspice, Multisim, ETAP, AutoCAD
- PCB & Hardware Design: PCB Design, PCB Layout, Schematic Capture, PCB Routing, Analog Circuit Design, Component Selection, Ground Plane Design, DRC, ERC, Gerber Generation, Bill of Materials (BOM), Manufacturing Documentation
- Automotive & EV Tools: CANalyzer, CANoe (basic), EV Benchmarking, VCU Interface Testing

EXTRACURRICULAR ACTIVITIES
- Handball — Left Wing Player
    - 1st Place - Intra-Campus Handball Competition, May'23
    - Runner-Up - Inter-Campus Handball Competition, Dec'22